06F — Deep-Dive Feature Audit: Pills & Alchemy (Part 6 of 6 — Grand Finale)

Master Holistic Overview of the Entire Pill, Toxicity & Alchemy Ecosystem: The 4-Pathway Split-Brain, `applyQualityPill` Blindness, Dual-ID Stock Fragmentation, Visiting Medicine Merchants & Unified Architecture

- **Target Codebase**: `JianghuSect 2` (`JS2-v2.6.0` in Google Drive `Js2 new`)
- **Target Audience**: Autonomous AI Developers (**GPT** & **Gemini**)
- **Role of Document `06F`**: **Master Holistic Synthesis** concluding the 6-part Pills & Alchemy Deep Audit (`06A–06F`). Every cross-module seam between `economy.js`, `cpeco.js`, `external_market.js`, `bottleneck.js`, `stats.js`, `cultivation.js`, `combat.js`, `content.js`, `fishing.js`, `sect.js`, and `sim.js` is synthesized into a single unified architecture.
- **Immutable Project Rules**:
  1. **NEVER modify `js/human_core.js`** (`HC.FINGERPRINT === '155:2f9c9dd7'`).
  2. **Preserve exact 40 `<script>` tags** and load order in `index.html` / `test/load.js`.
  3. **Strict Closed-Loop Economy & Deterministic Save/Load**.

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1. `[PILL-SYS-01]` Architectural Root Cause: The 4-Pathway Pill Consumption Split-Brain & Why Pill Quality (`q`) Does `0%` on `80%` of Pill Effects!

Across `JS2-v2.6.0`, consuming a pill passes through **four divergent code paths** wrapped across four separate files (`economy.js` $\to$ `stats.js` $\to$ `cpeco.js` $\to$ `bottleneck.js`):

1. `ECO.applyPill(W, npc, key)` — monkey-patched 3 times (`economy.js:338` $\to$ `stats.js:658` $\to$ `cpeco.js:395`).
2. `ECO.applyQualityPill(W, npc, key, quality)` (`economy.js:369–376`).
3. `ECO.usePill(W, npc, key)` — monkey-patched twice (`economy.js:377` $\to$ `bottleneck.js:168`).

4. `ECO.useConsumableItem(W, npc, itemId, opts)` (`economy.js:1312–1385`).

Three Systemic Bugs Inside `ECO.applyQualityPill` (`js/economy.js:369–376`):

```JavaScript
ECO.applyQualityPill = (W, npc, key, quality) => {
  const s = npc.h.state,
        before = { hp: s.hp, qi: s.qi, energy: s.energy, rest: s.rest, heartDemon:
s.heartDemon },
        result = ECO.applyPill(W, npc, key);
  if (!result) return result;
  const q = clamp(Number.isFinite(quality) ? quality : 1, 0.7, 1.5);
  for (const k of Object.keys(before))
    if (Number.isFinite(before[k]) && Number.isFinite(s[k])) s[k] = before[k] +
(s[k] - before[k]) * q;
```

1. **`80%` of Pill Effects Completely Ignore Pill Quality `q`!**:
   - Because `applyQualityPill` only diffs `{ hp, qi, energy, rest, heartDemon }`, **NONE of the following scale with `q`**:
     - `cultivationProgress` (`CULT.gain`) — `q = 1.65` gives the **exact same** exp as `q = 0.70`!
     - `rateBoost` (`c.pillBoost`), `breakthroughBonus` (`c.btBonus`), `clearToxic` (`c.toxic`), `healInjury` (`npc.inj`), `lifespanBonus` (`c.lifeBonus`), `restoreBeastFatigue`, and `cbBuff`/`grantBuff` potency!
   - **Toxicity (`c.toxic`) Never Scales Inversely With Purity**: In Xianxia alchemy lore (丹毒), crude low-purity pills (`q = 0.72`) leave heavy toxic dregs (`1.30×` toxicity), whereas Supreme Nine-Vein Pills (`q >= 1.55`) are crystal-pure (`0.50×` toxicity)! Currently a `1.65` Masterpiece Pill poisons meridians identically to `0.70` burnt slag—rendering Grandmaster `ALCHEMY` skill, `PEAK_DIVINE_CAULDRON`, and Mithril Cauldrons largely cosmetic!
2. **Hard `1.50` Quality Clamp Deletes `1.50..1.77` Masterwork Quality (`economy.js:108, 366, 368, 372` vs `261`)!**:
   - `ECO.finishCraft` (`line 261`) brews pills up to **`q = 1.77`** (`1.65 + kwPrecBonus 0.12`), **yet `ECO.add`, `ECO.pillQuality`, `ECO.addPersonalPill`, and `ECO.applyQualityPill` all clamp `q` down to `1.50`**, erasing Peak & Craft-Precision bonuses above `1.50`! Raise the pill quality ceiling consistently to `1.80`.
3. **Pre-Clamp Truncation Bug on Low-Quality Heals (`line 373`)**:
   - `applyPill` clamps `s.hp` to `hpMax` *before* `applyQualityPill` multiplies `(s.hp - before.hp) * q`! At `70/100` HP, a `q = 0.75` `P:BASIC_HP2` (`raw +84`, `scaled +63` $\to$ should reach `100/100`) gets clamped to `delta = 30`

first, then scaled to `70 + 30 * 0.75 = 92.5`! Pass `q` **into** the core effect calculation before clamping!

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2. `[PILL-SYS-02]` Dual-Key Stock Fragmentation (`'P:' + recipeId` vs `r.outItemId 'ITEM190_PILL_*'`) Splits Stock, Scarcity & Merchant Trade!

Compare how `ECO.finishCraft` (`js/economy.js:265`) stores brewed pills vs how `external_market.js:126, 169` generates merchant stock/wants:

- **In `ECO.finishCraft` (`economy.js:265`)**: `const key = r.outItemId ? r.outItemId : 'P:' + r.id;` $\to$ Brewed `REC190_PILL_002` enters `W.sect.stock` as **`'ITEM190_PILL_002'`**.
- **In `external_market.js:126–129` (`buildStock`) & `169–172` (`buildWants`)**: `Object.keys(ECO.recipeById).filter(...).map(rid => 'P:' + rid)` $\to$ Visiting Merchants buy and sell **`'P:REC190_PILL_002'`**!
- **Three Cascading Failures**:
  1. **Cannot Sell Surplus Brewed Specialty Pills to Medicine Merchants**: When a merchant wants `'P:REC190_PILL_002'`, `ECO.count(W, 'P:REC190_PILL_002')` is **`0`** even when `W.sect.stock['ITEM190_PILL_002'] === 25`!
  2. **Duplicate Split Piles in `W.sect.stock` & `n.inv.pills`**: Buying `'P:REC190_PILL_002'` from a merchant creates a second separate key alongside brewed `'ITEM190_PILL_002'`, splitting `CPE.scarcity`, `ECO.inventoryIntelligence`, `pillQuality`, and `have(k)` in `CPE.pickPurchase` (causing disciples to buy both keys thinking they are different pills)!
  3. **Exact Fix**: Introduce **`ECO.canonicalPillKey(id)`**: if `id.startsWith('P:')` and `ECO.recipeById[id.slice(2)]?.outItemId`, resolve to `r.outItemId` (unless `BASIC_*`/`BT_*`/`CB_*` which have no `outItemId`, keeping `'P:BASIC_HP'`, `'P:BT_XIAN_1'`, `'P:CB_FURY'` 100% unchanged for backward test compatibility!), and use `canonicalPillKey` in `external_market.js` (`buildStock` / `buildWants`)!

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3. [PILL-SYS-03] Visiting Medicine & Rare Merchants (external_market.js:230, 378–399) — Sect & Disciples Are Blind to Breakthrough, Longevity, Insight & Combat Pills!

Inspect `pillNeed(n, id)` in `js/external_market.js:378–387`:

```javascript
function pillNeed(n, id) {
  const info = ECO.pillInfo(id); if (!info) return 0;
  const e = info.effect || {}, s = n.h.state, hm = Math.max(1, CULT.hpMax(n));
  if (e.healInjury && n.inj && n.inj.length) return ...;
  if (e.healHealth && s.hp < hm * 0.72) return ...;
  if (e.qiRestore && n.cult && s.qi < CULT.qiMax(n) * 0.25) return ...;
  if (e.clearToxic && n.cult && (n.cult.toxic || 0) > 25) return ...;
  if (e.clearHeartDemon && s.heartDemon > 35) return ...;
  if (e.energyRestore && s.energy < 25) return 0.6;
  return 0; // <-- RETURNS 0 FOR ALL BREAKTHROUGH, LONGEVITY, INSIGHT, BEAST &
COMBAT PILLS!!
}
```

- **Major Emergent Market Blindness**:
  1. When *Immortal Alchemist Lin (เซียนโอสถหลิน)* or *Auction Broker Jin (นายหน้าหอประมูลจิน)* visits carrying a **`P:BT_*` Breakthrough Pill** needed by a bottlenecked Elder, a **`REC140_ALC_LONGEVITY_6` Immortal Peach Pill** needed by a dying Elder, or a **`P:CB_*` War Pill** during heightened threat, `pillNeed` returns **`0`**! Neither `EXT.tradeSect` (`line 230`) nor wealthy Elders (`EXT.memberQuote line 395`) recognize its value!
  2. Furthermore, when `EXT.tradeSect` (`line 301`) or `EXT.tradeMember` (`line 489`) buys pills from a merchant, `pillQuality` is never initialized (`defaults to 1.0`, whereas `'rare'` / `'medicine'` merchants should offer `q = 1.05..1.30` specialty batches), and `EXT.memberQuote` (`line 390`) never checks `CULT.isAbstainingFromPills(n)`!

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4. [PILL-SYS-04] Pill-Poison Lifecycle (c.toxic), Symbiotic VENOM_GU Digestion, ELIXIR_DAO Constitution & Death/Exile Pill Loss!

2. **Missing Pill-Toxicity Overdose Backlash (`c.toxic >= 90`)**:

- Currently `c.toxic` caps silently at `100` with zero acute danger. When an NPC recklessly gorges pills up to `c.toxic >= 90` (พิษโอสถอุดตันเส้นชีพจร), trigger a Meridian Toxin Backlash (`poison` status + minor meridian injury + `+6 heartDemon`), strongly motivating high-toxicity cultivators to seek `P:BASIC_DETOX`, `SERENE_PURIFICATION` mantra stance (`03C`), `ALCHEMICAL_INCENSE_MIST` support aura (`03F`), or `spirit_pond` cleansing!

2. **Path-Specific Pill Metabolism (`VENOM_GU` & `ELIXIR_DAO` / `PILL_FURNACE_IMMORTAL`)**:
   - **`VENOM_GU`**: Passively digest `0.8` `c.toxic`/hour into `+1.5 SYMBIOTIC_GU` gauge and `+0.4` cultivation progress (เลี้ยงหนอนกู่ด้วยพิษโอสถ)!
   - **`XIAN.ELIXIR_DAO` & `CRAFT_CREATION.PILL_FURNACE_IMMORTAL`**: Gain `-35%` pill toxicity accumulation and `+12%` pill absorption efficiency (กายาหม้อหลอมโอสถ)!
3. **Personal Pills Vanish on Death/Exile & Surplus `n.inv.pills` Cannot Be Donated (`ai.js`, `sim.js`, `cpeco.js:209`)**:
   - In `AI.kill` and `SIM.retire` (`non-violent retirement/expulsion`), rescue `n.inv.pills` and `n.inv.craftedPills` back into `W.sect.stock` (or bequeath to personal `apprentice`), instead of letting priceless Tier-6/7 pills vanish with dead records!
   - In `A.donate` (`cpeco.js:209`), allow disciples to donate surplus/obsolete `n.inv.pills` (`e.g. outgrown P:BT_* pills or held > 3`) via the already-implemented `ECO.donatePersonalProduct(W, n, pillKey, qty)` (`economy.js:315`)!

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5. Unified Single-Authority Pill Engine (`ECO.consumePillUnified`) & Master `P0` $\to$ `P3` Roadmap (`06A–06F`)

Replace the fragmented 4-pathway monkey-patches with **one clean internal core `ECO.consumePillCore(W, npc, rawKey, opts)`** (keeping `ECO.applyPill`, `ECO.applyQualityPill`, `ECO.usePill`, and `ECO.useConsumableItem` as thin backward-compatible facades calling `consumePillCore`):

Phase	Priority	Target Files	Exact Deliverables Across `06A–06F`
Phase 1	**P0 — Critical Broken Mechanics & Vow Shattering**	`economy.js`, `cpeco.js`, `bottleneck.js`,	1) Enforce `CULT.isAbstainingFromPills(n)`

Phase	Priority	Target Files	Exact Deliverables Across `06A–06F`
		`stats.js`, `sim.js`, `sect.js`	(`daoOath` + `VOW_NO_MEDICINE`) across `SIM.hourly`, `CPE.*`, `BN.prepare` (`bottleneck.js:146`), `ST.preFight`, and `EXT.memberQuote` (`06A/06C/06D`). 2) Fix `qiRestore` `100×` unit mismatch (`> 1.5 ? / 100`), move `qiRestore`/`energyRestore`/`restRestore` into base `economy.js`, and replace ghost `'P:HEAL'` with `'P:BASIC_HP'`/`'P:BASIC_WOUND'` (`06A`). 3) Unify Phase-2 queue onto `npc.ext.multiPhasePills` (deleting dead `pendingPillPhases`), fix double Phase-1 payout in `useConsumableItem`, fix swapped `dur` arg in `tickMultiPhasePills`, and fix `'YIN_SHADOW'` zero-tox check (`06E`).

Phase	Priority	Target Files	Exact Deliverables Across `06A–06F`
Phase 2	**P1 — Breakthrough, Quality Scaling & Combat Pre-Fight Fixes**	`bottleneck.js`, `economy.js`, `stats.js`, `data_items.js`, `cultivation.js`	1) Route personal-bag `P:BT_*` pills in `BN.prepare` through `BN.pillMult` (`btTaken++`, `q` scaling, `×1.8` wrong-stage/path toxicity) & reset `_btBonusPct` on `BN.onResult` (`06C`). 2) Make `applyQualityPill` (`q` up to `1.80`, applied *before* HP/Qi clamp) scale `cultivationProgress`, `btBonus`, `clearToxic`, `healInjury`, `lifespanBonus`, `cbBuff`, and **inversely scale `toxicity`** (`06F`). 3) Fix `ST.preFight` (`stats.js:663`) so `professionMeal` food buffs no longer block `P:CB_*` war pills, block free stock-raiding in spars/`away` fights, and deduct `lifespanCostYears: 4` + enforce `forbiddenVault` on

Phase	Priority	Target Files	Exact Deliverables Across 06A–06F
			ITEM190_PILL_012 (06D).
Phase 3	**P2 — 43-Path Bottleneck Parity, EFX_* Wiring & Herb Protection**	data_bt.js, bottleneck.js, economy.js, cultivation.js	1) Provide Lineage/43-Path P:BT_* & BT.TREASURES coverage + allow CPE.pickPurchase & all ranks to use general breakthrough pills (06C). 2) Bridge all 20+ unmapped EFX_* effects into ECO.pillInfo/applyPill; persist npc.ext.dantianBonusPct into CULT.qiMax; unify npc.ext.pillLifespanYears with diminishing returns; fix starter: true double-yield on Tier 5–7 recipes (06B). 3) Enforce recipeTier/ALCHEMY skill & tier ASC + reserveFloor herb selection in brewMultiPhaseSpecialtyPill/consumeSectHerbs + close 20-stone world-pool leak (06E).

Phase	Priority	Target Files	Exact Deliverables Across `06A–06F`
Phase 4	**P3 — Market Intelligence, Specialty Recovery, Toxicity Lore & Inheritance**	`external_market.js`, `cpeco.js`, `ai.js`, `content.js`, `fishing.js`	1) Canonicalize `'P:' + rid` $\leftrightarrow$ `outItemId` in `external_market.js` & expand `EXT.pillNeed` to Breakthrough, Longevity, Insight, Beast & Combat pills (`06F`). 2) Restore `ITEM190_PILL_001..004, 008..011` specialty behaviors (`clearPoison`/Gu cleanse, `maxRecipients: 4` party heal, `curesMeridianWound`, `restoreBeastFatigue` AI, `shadowVeil` scout AI) (`06A/06D`). 3) Add `c.toxic >= 90` overdose backlash, `VENOM_GU` toxin digestion, `ELIXIR_DAO` resistance, Spirit-Fish alchemy reserves, physical `researchAlchemy` act, and death/surplus personal pill recovery (`06B/06E/06F`).